**Response Letter with Annotations**

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences, 120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120

**Annotations**

1. **Purposes**
   - **a.** Motivation: Regardless of the specific approach used for these tasks (supervised, semisupervised, or unsupervised), labeled data are needed to build a training set or a gold standard against which performance can be assessed.
   - **b.** Specific Rationale (several bullet points)
     - Prior strategies focused on:
       1. Recruiting and training coders, such as research assistants.
       2. Relying on crowd workers on platforms such as Amazon Mechanical Turk (MTurk).
       3. Combine these two strategies: trained annotators create a relatively small gold standard dataset, and crowd workers are employed to increase the volume of labeled data.
     - Concerns raised:
       1. Trained coders involve a high cost
       2. With crowd workers, quality may be insufficient, particularly for complex tasks and languages other than English
     - Alternative options:
       1. Potential of large language models (LLMs) for text-annotation tasks, with a focus on ChatGPT.
       2. ChatGPT is significantly cheaper than MTurk. ChatGPT's per-annotation cost is about $0.003 or a third of a cent—about thirty times cheaper than MTurk, with higher quality.
   - **c.** Research Questions and/or Hypotheses
     - Does ChatGPT outperform crowd workers in text annotation tasks?
     - Can large language models (LLMs) provide higher-quality and more cost-efficient annotations?
     - **Identify IV, DV, mediator, moderator**
       1. **IV:** Type of annotator (ChatGPT vs. crowd workers vs. trained annotators)
       2. **DV:** Accuracy and intercoder agreement
     - Identify specific relationships among variables
       1. ChatGPT → higher accuracy than crowd workers
       2. ChatGPT → higher intercoder agreement than both crowd workers and trained annotators

2. **Method**
   a. Research Design
      - Computational study
   b. Sample and Data Collection
      - Sample size was determined by the number of texts needed to build a training set for a machine learning classifier.
      - The analysis relies on four datasets:
        1. A random sample of 2,382 tweets drawn from a dataset of 2.6 million tweets on content moderation posted from January 2020 to April 2021
        2. A random sample of 1,856 tweets posted by members of the US Congress from 2017 to 2022, drawn from a dataset of 20 million tweets
        3. A random sample of 1,606 newspaper articles on content moderation published from January 2020 to April 2021, drawn from a dataset of 980k articles collected via LexisNexis.
        4. Replicated the data collection for (i), but for January 2023.
   c. Data collection
      - Trained three political science students to conduct the annotation tasks.
      - Restricted access to the tasks to workers who are classified as "MTurk Masters" by Amazon, who have an HIT (Human Intelligence Task) approval rate greater than 90% with at least 50 approved HITs, and who are located in the United States.
      - ChatGPT API with the "gpt-3.5-turbo". The annotations were conducted between March 9–20 and April 27–May 4, 2023.
        1. Avoided adding any ChatGPT-specific prompts to ensure comparability between ChatGPT and MTurk crowd workers
        2. Temperature parameter at 1 (default value) and 0.2 (which makes the output more deterministic; higher values make the output more random).
        3. For each temperature setting, we collected two responses from ChatGPT to compute the intercoder agreement.
        4. Created a new chat session for every tweet to ensure that the ChatGPT results are not influenced by the history of annotations.
   d. Measures
      - **Annotation tasks**
        1. Relevance: whether a tweet is about content moderation or, in a separate task, about politics
        2. Topic detection: whether a tweet is about a set of six predefined topics (i.e., Section 230, Trump Ban, Complaint, Platform Policies, Twitter Support, and others.
        3. Stance detection: whether a tweet is in favor of, against, or neutral about repealing Section 230 (a piece of US legislation central to content moderation)
        4. General frame detection: whether a tweet contains a set of two opposing frames ("problem" and "solution"). The solution frames

content moderation as a solution to other issues (e.g., hate speech). The problem frame describes tweets framing content moderation as a problem on its own, as well as in relation to other issues (e.g., free speech)

5. Policy frame detection: whether a tweet contains a set of fourteen policy frames proposed

- **Accuracy**: measured as the percentage of correct annotations (using our trained annotators as a benchmark),
- **Intercoder agreement**: computed as the percentage of tweets that were assigned the same label by two different annotators (research assistant, crowd workers, or ChatGPT runs).

**e.** Analysis

- Computed average accuracy (i.e., percentage of correct predictions), that is, the number of correctly classified instances over the total number of cases to be classified, using trained human annotations as our gold standard and considering only texts that both annotators agreed upon.
- Second, intercoder agreement refers to the percentage of instances for which both annotators in a given group report the same class
- Pearson's correlation

**3. Results and Conclusions**

**a.** Answers to RQs and Hypotheses

- ChatGPT outperforms MTurk for most tasks across the four datasets. On average, ChatGPT's accuracy exceeds that of MTurk by about 25 percentage points
- ChatGPT demonstrates adequate accuracy overall, considering the challenging tasks, number of classes, and zero-shot annotations.
- ChatGPT's performance is very high. On average, intercoder agreement is about 56% for MTurk, 79% for trained annotators, 91% for ChatGPT with temperature = 1, and 97% for ChatGPT with temperature = 0.2.
- The correlation between intercoder agreement and accuracy is positive (Pearson's r = 0.36).
- ChatGPT's accuracy is positively correlated with the intercoder agreement of trained annotators (Pearson's r =0.46), suggesting better performance for easier tasks.
- Conversely, ChatGPT's outperformance of MTurk is negatively correlated with the intercoder agreement of trained annotators (Pearson's r = -0.37), potentially indicating stronger overperformance formore complex tasks.

**b.** Include Key Figures/Tables

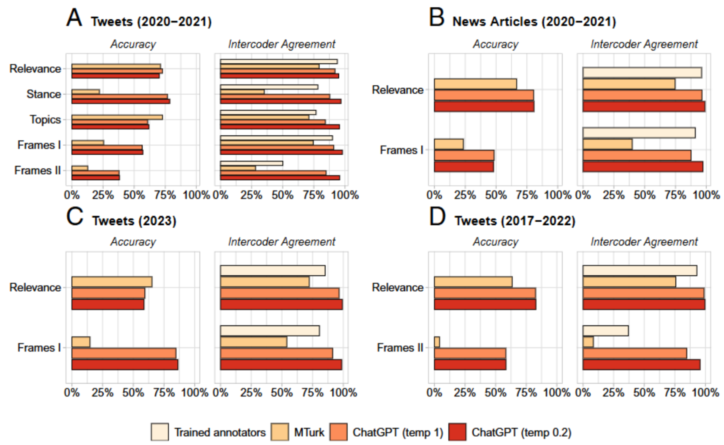


**Fig. 1.** ChatGPT zero-shot text annotation performance in four datasets (A: tweets, 2020-2021; B: news articles, 2020-2021; C: tweets, 2023; D: tweets, 2017-2022), compared to MTurk and trained annotators. ChatGPT's accuracy outperforms that of MTurk for most tasks. ChatGPT's intercoder agreement outperforms that of both MTurk and trained annotators in all tasks. Accuracy means agreement with the trained annotators.

**c.** Authors' Conclusions

- ChatGPT's performance is impressive, particularly considering that its annotations are zero-shot.

## 4. Discussion

**a.** Implications

- This paper demonstrates the potential of LLMs to transform text-annotation procedures for a variety of tasks common to many research projects.
- The evidence is consistent across different types of texts and time periods.
- Strongly suggests that ChatGPT may already be a superior approach compared to crowd annotations on platforms such as MTurk.
- The following questions seem particularly promising: i) performance across multiple languages; ii) implementation of few-shot learning; iii) construction of semiautomated data labeling systems in which a model learns from human annotations and then recommends labeling procedures (12); iv) using chain of thought prompting and other strategies to increase the performance of zero-shot reasoning (13); and v) comparison across different types of LLMs.

**b.** Limitations

- We underscore that the test to which we subjected ChatGPT is hard. Our tasks were originally conducted in the context of a previous study and required considerable resources.
- Developed most of the conceptual categories for our particular research purposes. Moreover, some of the tasks involve a large number of classes and exhibit lower levels of intercoder agreement, which indicates a higher degree of annotation difficulty.

**c.** Future Directions

- At the very least, the findings demonstrate the importance of studying the text-annotation properties and capabilities of LLMs more in depth.

- Further research is needed to better understand how ChatGPT and other LLMs perform in a broader range of contexts.

**--- Page Break Here ---**

## Response Letter (~600 words)

### Assessment of the study

Gilardi et al. (2023) analyze a very timely evaluation of LLMs, specifically in the context of text annotation tasks. I believe the study addresses a central challenge in computational social science on the need for high-quality labeled data. As the authors noted, researchers either employ trained coders or crowd workers, or both. This involves certain trade-offs between cost and quality. The analysis of ChatGPT as a potential strategy helps reduce these trade-offs, as they have shown to produce high-quality results and is cost-effective. This study highlights the evolving role of artificial intelligence in research.

### Strengths

One strength of the paper lies in its clear and systematic comparative design. Although the paper is short, it is clear in what it seeks to achieve. By evaluating ChatGPT against both crowd workers and trained annotators across multiple datasets and tasks, the authors provide evidence of its performance. The analysis of cost when using these platforms adds an important practical dimension, highlighting not only the performance of ChatGPT but also its efficiency.

Another strength lies in the scale and diversity of the datasets. The study includes over 6,000 texts across different time periods and formats, from tweets to news articles, which enhances the reliability of the findings. This shows that ChatGPT can be used to analyze diverse texts. The authors also test multiple types of annotation tasks, including relevance, stance, topic classification, and frame detection. This strengthens the argument that ChatGPT's performance is not limited to a narrow set of conditions.

### Weaknesses

The reliance on trained annotators as a "gold standard" assumes that human judgments are correct and unbiased. This may not always be the case, especially for complex tasks such as frame detection. I say this because we are currently working on a study on how the media in Ghana frames an environmental issue called "galamsey." We are three coders, and we seem not to reach a consensus on which frames are used in the articles. Additionally, while ChatGPT demonstrates high performance, its outputs are generated through a "black box" process that lacks transparency. Unlike human annotators, whose reasoning can be examined, it is difficult to interpret how ChatGPT arrives at its classification, even if a prompt has been fed into it. This raises questions about reliability, accountability, and reproducibility.

Additionally, the study focuses primarily on English-language political content related to content moderation. I wonder if the findings would hold for other domains, languages, or more complex annotation tasks.

**Other notes**

Aside from these weaknesses, the article helps us to reflect on which annotational strategies to employ. Each strategy comes with its advantages and disadvantages, and it is always good for us to focus on the objectives of our study and choose which strategy works best. Finances also play a huge role in what strategy to employ, and, like the authors pointed out, LLMs can help us get quality and affordable data.